

with $k - 1$ punctures. We assume that both metrics are nondegenerate, that both metrics have one end with a common asymptotic necksize of ε_* , and that (M_1, g_1) admits a Jacobi field deforming the necksize ε_* to first order. We then glue the two metrics together end-to-end, along the end with the common asymptotic necksize of ε_* . Within this construction we have the freedom to rotate (M_1, g_1) relative to (M_2, g_2) , using any element of $SO(n - 1)$ fixing the common axis of the Delaunay asymptotes. We vary this rotation over the whole group, which then gives us a family of solutions in $\mathcal{M}_k$. In Section 7 we will complete the proof of Theorem 1.5 by showing the resulting submanifold cannot be contractible. The key tool we use is the forgetful map, sending a metric in $\mathcal{M}_k$ to its conformal class.

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2. PRELIMINARIES

In this section we recall some useful material from other sources and perform some preliminary computations.

2.1. Delaunay metrics. The Delaunay metrics are all the constant Q -curvature metrics on a twice-punctured sphere and, as we will see later, play an important role in understanding the behavior of singular constant Q -curvature metrics with isolated singularities.

Consider a metric $g = U^{\frac{4}{n-4}} \circ g$ on $\mathbf{S}^n \setminus \{p, q\}$ where p and q are distinct. After a rotation and a dilation, we can assume $p = N$ is the north pole and $q = S$ is the south pole. Transferring to the Euclidean gauge we write $g = u^{\frac{4}{n-4}} g_{\text{euc}}$ where $u : \mathbf{R}^n \setminus \{0\} \rightarrow (0, \infty)$. This metric has Q -curvature equal to $\frac{n(n^2-4)}{8}$ if and only if u satisfies (1.6) with a single singular point at the origin.

Frank and König [5] classified all the solutions of

$$u : \mathbf{R}^n \setminus \{0\} \rightarrow (0, \infty), \quad \Delta^2 u = \frac{n(n-4)(n^2-4)}{16} u^{\frac{n+4}{n-4}}, \quad (2.1)$$

and we describe them here. First we perform the Emden-Fowler change of coordinates, defining

$$\mathfrak{F} : \mathcal{C}^\infty(\mathbf{B}_r(0) \setminus \{0\}) \rightarrow \mathcal{C}^\infty((-\log r, \infty) \times \mathbf{S}^{n-1}), \quad \mathfrak{F}(u)(t, \theta) = e^{\frac{4-n}{2}t} u(e^{-t}\theta). \quad (2.2)$$

We can of course invert $\mathfrak{F}$, obtaining

$$\mathfrak{F}^{-1}(v)(x) = |x|^{\frac{4-n}{2}} v(-\log |x|, \theta).$$